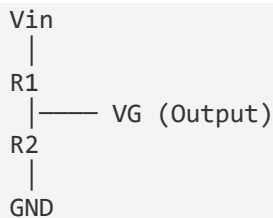


Understanding the Voltage Divider: Theory, Limitations & Applications

1 R1, R2, and VG — How They Relate

Every voltage divider question comes back to one formula. Get comfortable with this, and the rest is just logic.

$$VG = Vin \times R2 / (R1 + R2)$$



RULE 1

R2 and VG move in the SAME direction. Increase R2 → VG goes up. Decrease R2 → VG goes down. (VG is measured across R2, so whatever R2 claims, VG gets.)

RULE 2

R1 and VG move in OPPOSITE directions. Increase R1 → VG goes down. Decrease R1 → VG goes up. (R1 sits between the source and the output — anything it uses up never arrives at VG.)

Why This Happens — Ohm's Law

R1 and R2 are in series, so the exact same current flows through both. Since $V = I \times R$, and current is fixed, whichever resistor is larger simply drops more voltage across itself. No exceptions.

$$V(R1) = I \times R1 \quad V(R2) = I \times R2 \quad V(R1) + V(R2) = Vin$$

REMEMBER THIS ONE LINE

The larger resistor always drops the larger voltage across itself. Since VG is taken across R2, growing R1 steals voltage away from VG — while growing R2 hands more voltage to VG.

See It in Numbers

Scenario	R1	R2	VG ($Vin = 12\text{ V}$)
Balanced	1 kΩ	1 kΩ	6 V
R1 doubled	2 kΩ	1 kΩ	4 V ↓

R2 doubled	1 kΩ	2 kΩ	8 V ↑
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2 Why It Can't Power a Circuit

A voltage divider clearly makes a lower voltage — so why not use it as a mini power supply? Two independent problems rule this out.

Problem 1: The Output Voltage Depends on the Load

The formula $V_G = V_{in} \times R_2 / (R_1 + R_2)$ only holds when nothing else is connected to the output. Connect a real load, and it sits in parallel with R_2 — so R_2 is no longer just R_2 , it becomes $R_2 \parallel R_{load}$:

$$R_{eq} = (R_2 \times R_{load}) / (R_2 + R_{load})$$

Condition	Effective R_2	V_G
No load	1000 Ω	6 V
100 Ω load connected	≈ 91 Ω	≈ 1 V

WHAT WENT WRONG

The output collapsed from 6 V to about 1 V — not because anything failed, but because the load itself changed the divider ratio. A real power supply must hold its output steady no matter how much current the load pulls. A resistor divider physically cannot do that.

Problem 2: It Wastes Power Continuously

Even with no load at all, current flows through R_1 and R_2 non-stop, straight from source to ground. That current turns into heat — power that never reaches your actual circuit.

$$I = V_{in} / (R_1 + R_2) = 12 / 2000 = 6 \text{ mA} \quad P_{wasted} = V_{in} \times I = 72 \text{ mW}$$

That's 72 mW burned permanently just to produce 6 V — and it only gets worse as the resistors are made smaller to resist loading effects. This power isn't lost in any abstract sense: it's dissipated as heat directly in R_1 and R_2 , which is why divider resistors can run warm even with no load connected at all.

VERDICT

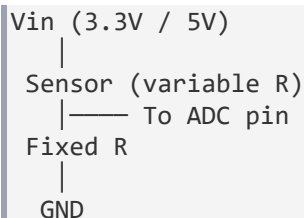
Unstable under load → continuous wasted power → zero short-circuit protection. This is exactly why real circuits use a linear regulator (7805, LM317) or a switching DC-DC converter — both use feedback to correct the output as the load changes, which a fixed resistor pair simply can't do.

3 Where Voltage Dividers Actually Get Used

A voltage divider isn't useless — it's just the wrong tool for delivering power. It's one of the most common building blocks for signals, sensing, and reference voltages, where little to no current needs to flow.

3.1 Reading Sensors

An LDR, thermistor, potentiometer, or flex sensor is just a variable resistor — it can't produce a voltage on its own, and an ADC can only read voltage. Put the sensor into a divider with a known fixed resistor, and its resistance change becomes a voltage change the ADC can read directly.



3.2 Logic-Level Shifting

A 5 V sensor output needs to talk to a 3.3 V microcontroller pin. A divider scales it down safely — and works well here because a digital input pin draws almost no current, so it barely disturbs the ratio.

$$R1 = 1\text{ k}\Omega, R2 = 2\text{ k}\Omega \rightarrow 5V \times 2k / 3k \approx 3.3V$$

3.3 Bias & Reference Voltages

Biasing a MOSFET gate, setting a BJT's base voltage, generating a comparator reference, or centering an audio signal — all need a stable voltage point, not real current. A divider is the standard way to create one.

3.4 Op-Amp Feedback / Gain Setting

In a non-inverting amplifier, the feedback resistors form a divider running in reverse — their ratio sets the gain instead of stepping a fixed voltage down.

$$\text{Gain} = 1 + (R_f / R_{in})$$

3.5 Battery Voltage Monitoring

A 12 V battery is outside a 3.3 V ADC's range. A divider scales it down for safe monitoring — a measurement task, so the small loading effect barely matters if the divider resistors are kept small relative to the ADC's input impedance.

3.6 Signal Attenuation

In audio and RF stages, a divider simply knocks a signal down by a fixed ratio before it reaches a sensitive input — line-level audio into a mic input, or a signal into an oscilloscope probe.

Final Summary

A voltage divider is designed for voltage scaling and signal conditioning—not for supplying power to loads.